Radiometric Correction Model and Land Cover Classification of Snow-Covered Terrains for ICESat-2 Photon-Counting Lidar

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***Abstract*— The signal strength is a fundamental parameter in radiometric applications for satellite lidars. Different from full-waveform lidars, photon-counting lidars cannot record the returned signal strength but only respond to the presence of the photon event and may miss some returned photons due to the dead time effect, i.e., introduce radiometric distortion. Based on the lidar equation and the response mechanism of photon-counting detectors, we propose a radiometric correction model to remove the impact of the nonlinear response and dead time of detectors for the photon-counting lidar borne on ICESat-2. The returned signal photon number is corrected by the proposed model with respect to the photon event number per shot (PNPS) and surface slope derived from ATL03/ATL08 products. Then, the optical throughput calibration factor of ICESat-2 is obtained from ATL06 products over high Antarctic plateau where has given reflectance and clear atmosphere, which is generally equal to 0.52. The atmospheric attenuation induced by the molecular, cloud, and aerosol is calculated from ATL09 products. In addition, the corrected radiometric parameters including the calculated surface reflectance and apparent surface reflectance (ASR) are applied to classify land cover types along laser tracks over snow-covered terrains. The results indicate that the signal strength and calibration constant are reliable after corrections, but the atmospheric attenuation is sometimes inaccurate, which further influences the derived surface reflectance. In classifications, the overall accuracy and Kappa coefficient based on the corrected ASR can achieve the best classification results with 88.80% and 0.69. The proposed radiometric correction model is very essential to radiometric applications for photon-counting lidars such as ICESat-2, especially for data captured on ice and bare land with relatively high reflectance.**

***Index Terms*— Dead time effect, ICESat-2, land cover classification, photon-counting lidar, radiometric correction, signal strength.**

Manuscript received 16 July 2022; revised 23 November 2022; accepted 12 January 2023. Date of publication 16 January 2023; date of current version 26 January 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 41971302, in part by the Academy of Finland Projects under Grant 336145 and Grant 337656, in part by the Strategic Research Council Project under Grant 314312, in part by the Chinese Academy of Science under Grant 181811KYSB20160040 and Grant XDA22030202, in part by the Shanghai Science and Technology Foundations under Grant 18590712600, in part by the Jihua Laboratory under Grant X190211TE190, and in part by the Huawei under Grant 9424877. *(Corresponding author: Yue Ma.)*

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Digital Object Identifier 10.1109/TGRS.2023.3237273

I. Introduction

As ONE category of lidars, satellite laser altimeter is an active instrument by using a laser pulse to measure the range to the illuminated target. Combined with the satellite attitude and laser pointing information, laser footprints on the ground can be geolocated in a certain coordinate frame. A large number of successive footprints with the satellite's motion can generate a ground surface profile of a planet with a vertical accuracy of decimeters, even centimeters [1]. Owing to the superior performance of satellite laser altimeters, National Aeronautics and Space Administration (NASA) has developed various payloads for measuring the topographic map of Earth, Moon, and planetary. To name a few, the Geoscience Laser Altimeter System (GLAS) onboard the Ice, Cloud and land Elevation Satellite (ICESat), the Advance Topographic Laser Altimeter System (ATLAS) onboard ICESat-2, and the Global Ecosystem Dynamics Investigation (GEDI) lidar for Earth observations [2], [3], [4], Clementine lidar and Lunar Orbiter Laser Altimeter for Moon observations [5], [6], Mars Orbiter Laser Altimeter, and Mercury Laser Altimeter [7], [8].

According to the detection mode and detectors, satellite laser altimeters are mainly divided into two categories [9]. When the returned laser signal strength is sufficiently large (i.e., normally at least thousands of photons), a linear mode based on avalanche photodiodes (APD) or photomultiplier tubes (PMTs) is employed to convert the returned laser signal into an analog waveform. A photon-counting detection mode based on PMT or other photon-sensitivity detectors is used to output the discrete photon events if such signal strength is extremely weak (i.e., normally only few photons per laser shot). By extracting and interpreting the characteristics parameters of received waveforms (including the signal energy, peak position, and pulsewidth, skewness) recorded by full-waveform satellite laser altimeters, the geometric and radiometric features of the illuminated target can be exactly retrieved, such as the surface roughness, slope, and reflectance [10], [11], canopy height and biomass of the vegetation [12], [13], and variations in the height and thickness of the ice sheet and sea ice [14], [15], [16]. In particular, the surface reflectance or subsurface optical properties can be directly derived from the radiometric corrected echo signal strength through removing the attenuation due to the system and environment induced distortions [17], [18], [19], [20]. The derived surface reflectance can be applied for various

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applications ranging from object classification to identify the presence of the water ice on the lunar or Mercury surface [21], [22].

Unlike full-waveform laser altimeters, the photon-counting ATLAS only responds to the presence of the echo signal and records the time tag for every photon event. The ATLAS instrument splits the laser energy into six beams at the wavelength of 532 nm with a high repetition rate of 10 kHz and a small divergence angle of approximately 24 μrad [23]. The six beams are arranged into three beam pairs, and each pair comprises one strong beam and one weak beam with an energy ratio of 4:1. The cross-track distance between the two beams in a pair is approximately 90 m. The 16-element PMTs detect every strong beam to generate 16 independent outputs, while for each weak beam, the photons from every four PMTs are summed to yield one output. The detectors can gather both signal photons and noise photons to generate photon events. Many noise filtering algorithms have been developed to extract the captured signal photon events [24], [25], [26]. With high-density geolocations of signal photon events, ICESat-2 can provide an approximately continuous topographic profile with an interval of 0.7 m in the along-track direction. Such profiles make the ATLAS appropriate for measuring ice sheet elevation and sea ice thickness, monitoring the sea-level changes, generating terrain elevation models, estimating vegetation characteristics and water storage, and even conducting satellite derived bathymetry [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38].

To the best of our knowledge, there are only a few published literatures concerning the radiometric properties of the ATLAS. Lu et al. [39] evaluated and corrected the radiometric influence due to the after pulsing effect in receiver optics and photon-counting detectors of ICESat-2, which is very fundamental to further calculate the ocean optical parameters. Wang et al. [40] combined the ATLAS product and optical imagery to derive benthic reflectance in clean waters. Gibbons et al. [41] evaluated the ATLAS radiometric performances in terms of the captured photon number per shot (PNPS) obtained by averaging the signal photon events for nominally successive 200 laser shots over Antarctic and Greenland ice sheet. Neuenschwander et al. [42] applied such PNPS within 100-m along-track segments to provide the radiometric assessment of ATLAS over three types of the vegetated surfaces (including the boreal forest, temperate forest, and tropical forest). However, due to the nonlinear effect and dead time of photon-counting detectors, the PNPS is not completely equivalent to the returned laser signal strength.

Specifically, when the laser signal is extremely weak (with few photons), for a single laser shot measurement, the actual photon number is randomly sampled by using a Poisson random distribution with a mean value of the theoretically expected photon number. Then, the photon-counting detector responds the actual photons and only records the photons outside the dead time duration as photon events [43]. These photon events actually are the geolocated photons in ATLAS data products. As mentioned above, through accumulating and averaging the photon events of the successive laser shots, the calculated PNPS can effectively reduce the randomness. Nevertheless, the response mechanism of photon-counting detector would probably miss some photons due to the dead time effect especially when the signal strength is relatively large. Hence, a dead time correction factor should be introduced to compensate the radiometric deviation between the PNPS and the expected signal photon number, which is an essential issue for the entire radiation transfer for photon-counting lidars.

In fact, the PNPS is synthetically determined by the signal strength, pulsewidth, and dead time duration of detectors. Current radiometric correction introduced by the dead time only associates with the noise effect but not links with the received signal [44]. The dead time correction factor should be adjustable based on the captured PNPS, echo signal pulsewidth, and dead time duration. Otherwise, the corrected radiometric information of the illuminated target may be inaccurate. The specific objectives of this article are to 1) establish a model to recover the expected signal photon number as a function of the PNPS and pulsewidth, (2) retrieve the surface reflectance and apparent surface reflectance (ASR) using the corrected signal strength (i.e., the expected signal photon number rather than PNPS) and other data derived from ICESat-2 products, and (3) classify the snow-covered area and bare land based on the calculated surface reflectance and ASR. The rest of this article is organized as follows. Section II describes the study site and the used datasets, Section III clarifies the proposed method and flowchart, Section IV gives the results and discussions, and Section V concludes the summaries of this study.

II. Materials

A. Study Area

In this study, we utilize the ICESat-2 data over a steep mountainous area to validate the proposed algorithm. The study site is located at the north of Gilgit-Baltistan in Pakistan with the geolocation center at (36°25′ N, 74°57′ E). It is primarily composed of snow-capped mountains and deep valleys nearly all year round. Three laser tracks of the strong beams are selected when ICESat-2 passed through this site on June 7, 2021. The tracks have approximately identical lengths of approximately 270 km and are numbered by the ground track of the first strong beam on the far right with ATLAS oriented in the forward direction. The steep geographical gradient in the study site makes it well suitable for qualifying the performance of the proposed algorithm. The location of the study area and three ground tracks are illustrated in Fig. 1.

B. Datasets

1) ICESat-2 Lidar Data: The ICESat-2 mission provides 21 different data products with three levels named from ATL00 to ATL21 [3]. This study employs the ICESat-2 ATL03, ATL06, ATL08, and ATL09 datasets on June 7, 2021. The ATL03 product is the most significant Level-2 data product comprising the longitude, latitude, elevation, time tag, and

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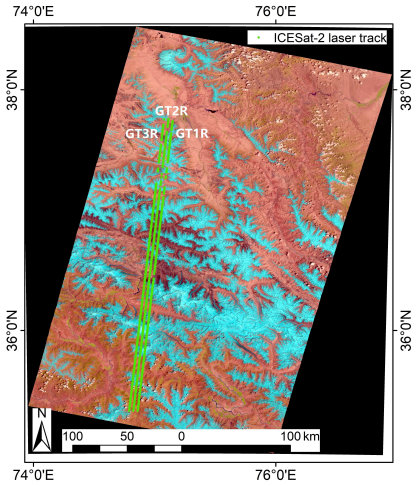


Fig. 1. Location of the study site and three ground tracks of ATLAS strong laser beams at the north of Gilgit-Baltistan in Pakistan. The ground tracks are illustrated by three green straight lines and correspond to GT1R, GT2R, and GT3R from right to left, respectively. The breaking points along each ground track represent the areas without valid photon events of ATLAS.

confidence level for each photon event and the transmitted energy measured by an internal laser energy monitor for each beam [45]. The ATL06 product gives the number of captured photon events and laser pulses within each 40-m along-track segment through the signal selection algorithm based on the ATL03 flags over land-ice area [46]. The ATL08 product determines the signal photon labels for the ground, canopy, and canopy top as well as the noise labels according to a differential, regressive, and Gaussian adaptive nearest neighbor method and data classified algorithm [47]. The ATL09 product is a Level-3A atmosphere product containing the calibrated and attenuated backscatter profiles and cloud characteristics parameters [48]. The ATL09 offers the molecular transmission profile from -1 to 20 km with 30-m resolution at 1 Hz and optical depth of particulates from cloud and aerosol at 25 Hz (approximately 280-m along-track resolution) to derive the atmospheric transmission.

2) Sentinel-2 Image Data: The Sentinel-2 optical instrument is a wide-swath, high-resolution, and multispectral imaging system with 13 spectral bands spanning from the visible to the near-infrared [49]. The Sentinel-2 teams release their reflectance products successively after the radiometric and geometric corrections. Sentinel-2 L1C product provides the top of atmosphere reflectance for all 13 spectral bands. To maintain temporal and spatial consistency with ICESat-2 lidar data and less than 10% cloud cover, three tiles on June 9, 2021 and one tile on June 4, 2021 in JPEG 2000 format from Sentinel-2 L1C products are selected. The normalized difference snow index (NDSI) data are usually regarded as the ground truth of snow-covered land, which can be calculated as the relative magnitude of the reflectance difference between visible (green) and shortwave infrared [50]. Thus, only the reflectance data for the third band with the central wavelength of 560.0 nm and the 11th band with the central wavelength of 1613.7 nm are employed to retrieve the NDSI in this study.

TABLE I
KEY ATLAS INSTRUMENT PARAMETERS OF THE TRANSMITTER AND RECEIVER

Parameters	Value
Transmitted laser wavelength	532 nm
Transmitted beam divergence (θ)	24 μrad (85% enclosed energy)
Transmitted laser energy (E_t)	~93 μJ (mean value of strong beams)
Transmitted laser pulse width (δ_t)	0.64 ns (root mean square width)
Effective aperture area of receiver (A)	0.41 m^2
Optical throughput efficiency of receiver (γ_r)	42%
Detector efficiency (γ_d)	15%

III. METHODS

A. Basic Lidar Equation for Spaceborne Lidars

The returned waveform captured by a satellite lidar is the temporal pulse signal generated by the propagation of the transmitted laser pulse through the atmospheric scattering and the reflection of the illuminated target. On the assumption that only the atmospheric attenuation effect is considered, the returned waveform from a Lambertian terrain with a uniform reflectance can be expressed by a Gaussian function as

$$n(t,t+\tau) = n_s(t,t+\tau) + n_n(t,t+\tau) = \int_t^{t+\tau} \frac{N_s}{\sqrt{2\pi}\delta_s} \exp\left(-\frac{t^2}{2\delta_s^2}\right) + f_n\tau \quad (1)$$

where δ_s and N_s represent the signal pulsewidth and the expected signal photon number, respectively; $n_s(t,\, t+\tau)$ and $n_n(t,\, t+\tau)$ correspond to the signal photon count and noise photon count within a time bin τ, respectively; and f_n is the noise rate. The expected signal photon number and pulsewidth of the returned signal associated with each transmitted laser pulse can be depicted by [51]

$$N_s = \frac{\gamma_r \gamma_d E_t}{E_p} \cdot \frac{A}{\pi R^2} \cdot T_a^2 \cdot \rho \cos\varphi \quad (2)$$

$$\delta_s = \sqrt{\delta_t^2 + \frac{4R^2 \tan^2\theta}{c^2}\left(\tan^2\theta + \tan^2\varphi\right)} \quad (3)$$

where γ_r is the optical efficiency of the lidar receiver, γ_d is the detector quantum efficiency, E_t is the transmitted laser energy that has considered optical efficiency of the transmitter, E_p is the energy of a single photon, A is the effective aperture area of the receiving telescope, R is the range from the lidar to the target, c is the light velocity, T_a is the one-way atmospheric transmittance, ρ is the surface reflectance of the measured target, δ_t is the transmitted laser pulsewidth, θ is the laser divergence angle, and φ is the laser incident angle concerning the surface slope and laser pointing angle. Some of key ATLAS instrument parameters are summarized in Table I [52].

The surface profile of terrains within a laser footprint is usually assumed to be a plane with roughness and slope (the influence of surface roughness is much weaker than surface slope due to the large footprint of satellite lidars), and such

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assumption nicely fits the footprint magnitude of ICESat-2 with an orbital height of 500 km and the divergence angle of 24 μrad. Hence, the laser incident angle can be expressed by using the surface slope and laser pointing angle as [51]

$$\varphi = \arctan\sqrt{\tan^2(\phi + s_{//}) + \frac{\tan^2 s_{\perp}\cos^2 s_{//}}{\cos^2(\phi + s_{//})}} \tag{4}$$

where ϕ is the laser pointing angle, and $s_{//}$ and $s_{\perp}$ are the along-track and across-track surface slopes, respectively. For ATLAS with a small pointing angle (i.e., <1.5°) [3], [52], laser incident angle φ can be approximated by the total surface slope (s_t) as

$$\varphi \approx s_t = \arctan\sqrt{\tan^2 s_{//} + \tan^2 s_{\perp}}. \tag{5}$$

Note that this approximation for ICESat-2 has a slight impact on estimating N_s especially when the surface slope is small, which will be quantitatively clarified in Section IV.

The expected signal photon number (N_s) is the direct measurement of the surface reflectance (ρ) when the specification parameters of a satellite lidar and the atmospheric transmittance (T_a) are exactly known. We can estimate the surface reflectance from (2) as

$$\rho = N_s \cdot \frac{\pi R^2 E_p}{\gamma_r \gamma_d E_t A} \cdot \frac{1}{T_a^2} \cdot \cos\varphi. \tag{6}$$

However, since the optical throughput of the lidar system may change during the flight campaign of a satellite lidar, a systematic calibration factor (C_F) is required, which can be determined by using data over the Antarctic plateau with the given surface reflectance and clear atmosphere [44]. Thus, a calibration constant C_s including all terms except the expected signal photon number, atmospheric transmittance, and surface slope is defined by

$$C_s = C_c \cdot C_F = \frac{\pi R^2 E_p}{\gamma_r \gamma_d E_t A} \cdot C_F \tag{7}$$

where C_c is a fixed hardware parameter and equals to 0.12 based on the ATLAS specification parameters in Table I. The variation of the calibration constant is mainly determined by the systematic calibration factor.

Hence, we can rewrite the surface reflectance of Lambertian terrains in (6) by a simplified formation as

$$\rho = N_s \cdot C_s \cdot \frac{1}{T_a^2} \cdot \cos\varphi. \tag{8}$$

It demonstrates that the surface reflectance strongly depends on the expected signal photon number N_s, the calibration constant C_s, the atmospheric transmittance T_a, and total surface slope φ.

B. Current Radiometric Correction Model for ICESat-2

The returned N_s is the key factor in the determination of the surface reflectance. Full-waveform lidars linearly respond and record the entire temporal profile of the returned signal (except for the saturation), and N_s can be usually calculated as the area under the signal waveform above a certain energy threshold. However, photon-counting lidars can only record the discrete time tag for each photon event and may miss some received photons due to the dead time effect. A Poisson distribution is normally used to represent the nonlinear response of photon-counting detectors, and the classic detection probability of the returned pulse signal considering the dead time and multiple channel detectors is formulated by [53], [54]

$$P_{sn}(K > 0) = \exp\left(-\frac{t_d f_n}{M}\right)\left[1 - \exp\left(-\frac{N_s + \omega f_n}{M}\right)\right] \tag{9}$$

where K denotes the average captured photon event number per detector, t_d is the dead time duration, M is the detector number, and ω is the signal duration (normally $\omega = 6\delta_s$).

Under the circumstance that no more than one signal photon is returned at one detector within the signal duration for a single laser shot, the detection probability can be described according to a Bernoulli trial by [55]

$$P_{sn}(K > 0) = \frac{N_p}{M} = \frac{N_t}{MN} \tag{10}$$

where N_p represents the captured PNPS in all detectors (e.g., $M = 16$ for strong beams of ICESat-2), and N_t is the total recorded photon event number for N laser shots. Thus, the expected signal photon number N_s can be expressed by substituting (10) into (9) as

$$N_s = -M\ln\left[1 - \frac{N_p}{M}\exp\left(\frac{t_d f_n}{M}\right)\right] - \omega f_n. \tag{11}$$

Given that both the detector dead time duration and the signal duration usually are less than tens of nanoseconds and the noise rate is normally megahertz magnitude, the expected signal photon number N_s can be simplified in terms of Taylor expansion method as

$$N_s = N_p\left(1 + \frac{1}{2M}N_p\right). \tag{12}$$

Here, the contribution of the higher order terms of the PNPS to the expected signal photon number N_s is ignored, and this simplified formation is similar to the dead time correction of (N_s) for ATLAS [44].

However, in (9) and (11), the signal duration ω and the dead time duration t_d are only associated with the noise rate f_n but not linked with the expected signal photon number N_s. For ICESat-2, the noise rate is normally several megahertz in the daytime, the time bin τ is 200 picoseconds, and both the dead time duration and the signal duration are normally several nanoseconds, which means that the expected noise photon number within t_d and ω is less than 0.01 count (i.e., $f_n.t_d < 0.01$ and $f_n.\omega < 0.01$) and the noise number within each time bin is approximately 0.001 count (i.e., $f_n.\tau = \sim 0.001$). When the ICESat-2 lidar receives a very weak signal (e.g., $N_s < 0.1$ counts per laser shot per detector and $N_s < 1.6$ counts for total 16 detectors) with ω of several nanoseconds, the average signal photon number within τ is less than 0.01 count [i.e., $N_s/(\omega/\tau) < 0.01$]. Generally, (9), (11), and (12) work well under a very weak signal level. In other words, within the dead time duration or signal duration, the signal count can hardly occur twice or more times; otherwise, (9) and (11) cannot work as they do not consider the dead time effect for signal photons. However, on bare-land and ice-covered areas

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(which are the main land-cover types in this study), N_s does not satisfy the very weak signal condition ($N_s < 1.6$ counts for total 16 detectors).

The noise rate f_n is stable or constant within a shot or even several shots, which evenly occurs within the entire range gate. This is the reason we can calculate the expected noise photon number within t_d and ω as $f_n.t_d$ and $f_n.\omega$ in (9) and (11). On the contrary, signal count $n_s(t,\ t+\tau)$ does not evenly occur, as shown in (1). If we consider the dead time effect for signal counts, the rigorous detection probability P_{sn} in (9) has to be derived by the recursive method [56]. Specifically, the detection probability of the first time bin is derived, then the detection probability of the second time bin, and then that of the ith time bin. For the PMT detectors, the detection probability of the ith time bin can be expressed as

$$P_{sn_i}(t_i) = \prod_{m=i-n_{\text{td}}}^{i-1} \left[1 - P_{sn_m}(t_m)\right] \times \left[1 - P_{sn_i-1}(t_{i-1}, t_{i-1}+\tau;\ K=0)\right] \quad (13)$$

where t_i is the time tag of the ith time bin, $n_{\text{td}} = t_{\text{d}}/\tau$ is the number of time bins that the dead time duration contains, and $P_{sn_0}(t_0, t_0+\tau;\ K>0)$ is defined as 0. Since the signal occurs randomly and unevenly, it would be prohibitively complex to enumerate all cases for the expectation of the recursive detection probability.

To complicate matters further, the pulse pileup effect leads to the changeable dead time duration for PMTs [43]. The dead time duration t_d is no longer a fixed value but varies with N_s and signal width δ_s, which can be expressed as [43]

$$\begin{aligned}
\bar{t}_d(\delta_s, N_s) &= P(t-3\delta_s, t+3\delta_s;\ K=1)\cdot w_0 \\
&\quad + P(t-3\delta_s, t+3\delta_s;\ K=2) \\
&\quad \cdot \int_{-3\delta_s}^{3\delta_s} \left(w_0 + t_{\text{op}}\right) p\left(w_0 + t_{\text{op}}\right) \mathrm{d}t_{\text{op}} \\
&\quad + P(t-3\delta_s, t+3\delta_s;\ K=3) \\
&\quad \cdot \int_{-3\delta_s}^{3\delta_s} \left(w_0 + t_{\text{op}}\right) p\left(w_0 + t_{\text{op}}\right) \mathrm{d}t_{\text{op}} + \cdots \\
&= w_0 n_{\text{PE}} e^{-n_{\text{PE}}} \\
&\quad + \sum_{k=2}^{+\infty} \left[\frac{n_{\text{PE}}^k}{k!} e^{-n_{\text{PE}}} \int_{-3\delta_s}^{3\delta_s} \left(w_0 + t_{\text{op}}\right) p\left(w_0 + t_{\text{op}}\right) \mathrm{d}t_{\text{op}} \right] \\
&= w_0 (N_s + 6 f_n \delta_s) e^{-(N_s + 6 f_n \delta_s)} \\
&\quad + \sum_{k=2}^{+\infty} \left[\frac{(N_s + 6 f_n \delta_s)^k}{k!} e^{-(N_s + 6 f_n \delta_s)} \right. \\
&\qquad \left. \times \int_{-3\delta_s}^{3\delta_s} \left(w_0 + t_{\text{op}}\right) p\left(w_0 + t_{\text{op}}\right) \mathrm{d}t_{\text{op}} \right] \qquad (14)
\end{aligned}$$

where w_0 is the time span of the output current (triggered by one photon), which is normally higher than the discrimination threshold, t_{op} is the extra time extension caused by following photons, $p(w_0 + t_{\text{op}})$ corresponds to the probability that the time span equals $w_0 + t_{\text{op}}$, and n_{PE} is the total number of both signal and noise photons within the time duration of the laser pulse ($\pm 3\delta_s$) and can be expressed as $n_{\text{PE}} = N_s + 6 f_n \delta_s$. The expected signal photon number should be corrected by considering the signal duration (ω) or signal duration (δ_s). However, it is nearly impossible to directly use an analytical expression to model the dead time effect for PMTs and also an analytical radiometric correction model.

C. Empirical Radiometric Correction Model for ICESat-2

To address this issue, we have to use the photon simulation algorithm to generate photon-counting lidar data by the recursive method [57] and then fit the quadratic model with respect to the expected signal photon number. We change possible total surface slopes and expected signal photon numbers to generate Gaussian waveforms with ergodic different pulse widths and amplitudes. For each generated Gaussian waveform with a specific pulsewidth and amplitude, the actual signal photon numbers are randomly sampled by using a Poisson random distribution and are then responded and recorded as photon events by the ICESat-2 used PMT detectors [43]. During the PMTs respond and record photon events, the dead time effect is involved. This random sampling for the given signal photon number and surface slope should be repeated for hundreds of times (i.e., for hundreds of laser shots), and the average captured photon event number per laser shot is then calculated as the PNPS. We plan to establish a quadratic regression model with a confidence level of 95% in the estimation of the relationship between the given expected signal photon number and the surface slope, and the PNPS as

$$N_s = \left(\mu_{p1} + \mu_{p2} N_p\right) N_p + (\mu_{s1} + \mu_{s2}\varphi)\varphi + \mu_c \quad (15)$$

where μ_{p1}, μ_{p2}, μ_{s1}, μ_{s2}, and μ_c are the fitting parameters of the regression model.

This empirical quadratic expression directly models the relationship among the total surface slope (φ), the expected signal photon number (N_s), and the PNPS (N_p). Here, we use the total surface slope instead of the dead time duration t_d and signal duration ω (or signal width δ_s). The reason is that, first, for a photon-counting lidar such as ATLAS with given system parameters, the signal width (δ_s) only varies with the total surface slope (φ), as shown in (3) and (5). Second, the dead time duration of PMTs t_d is determined by N_s and δ_s, as shown in (14). N_s has been included in (15), and δ_s is only determined by φ for ICESat-2 as mentioned above. Third, from the ICESat-2 data, we can directly estimate both the total surface slope (φ) and the PNPS (N_p), which is sufficient to recover the expected signal photon number (N_s). Note that (15) is generally useful for spaceborne photon-counting lidars with PMTs when measuring terrestrial surfaces. However, as the fitting parameters are not only related to environmental parameters but also system parameters of a given photon-counting lidar. As a result, the fitted parameters in this study will be only specific to the photon-counting lidar borne on ICESat-2.

D. Surface Reflectance Retrieval and Land Cover Classification

Based on the radiometric correction model of illuminated terrains in (15) and the ICESat-2 data products, we further provide the detailed flowchart of the proposed algorithm on

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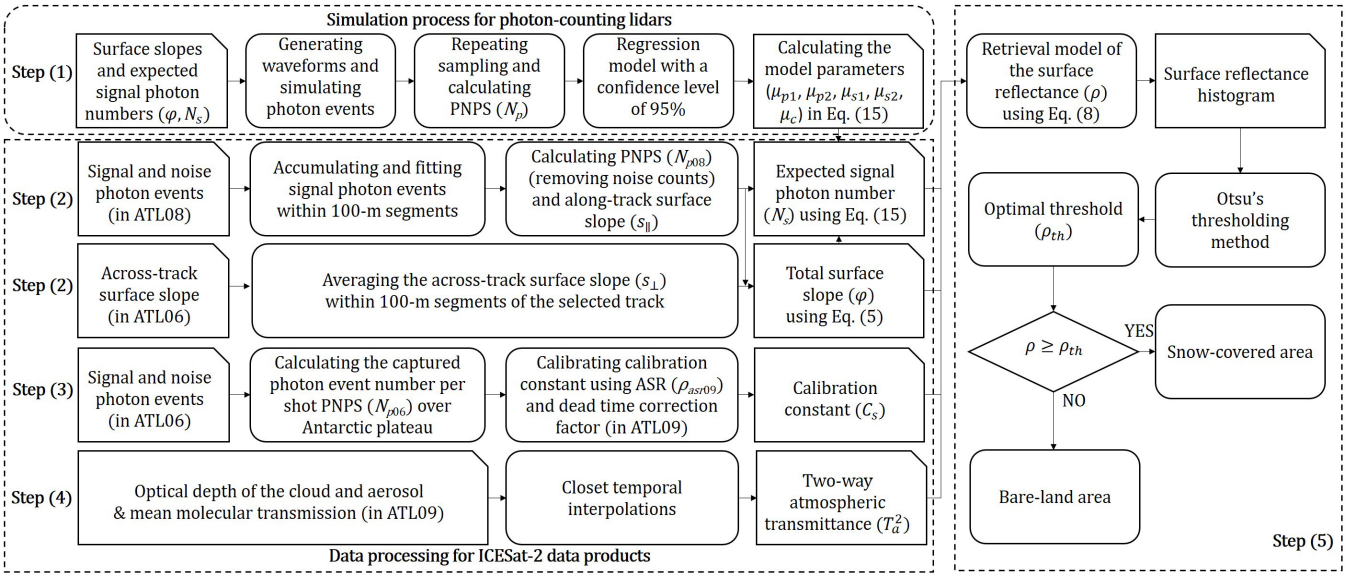


Fig. 2. Flowchart of the proposed algorithm on the retrieval of the terrestrial surface reflectance and classification of bare land and snow based on the ICESat-2 photon data.

the retrieval of surface reflectance and the classification of bare land and snow, as illustrated in Fig. 2.

Step (1): Based on the abovementioned photon event simulation method, the sampling photon events are accumulated to generate the PNPS. Then, the fitting parameters of the quadratic regression model with respect to the expected signal photon number (N_s) in empirical (15) can be estimated for correcting the dead time losses.

Step (2): We acquire the captured signal photon events within a 100-m along-track window for each ATL08 segment to calculate the PNPS (N_{p08}). Since that such PNPS is not adjusted for background noise [42], we remove the impact of the noise photons on the PNPS by considering the background count rate (bckgrd_rate, N_A) in the corresponding ATL03 product and signal integration height (h_canopy, h_s) in the ATL08 product as N_{P08}-$h_s N_A$. In addition, the along-track surface slope ($s_{//}$) can be directly extracted by a linear fitting of the labeled signal photons within each 100-m segment. The across-track surface slope ($s_{\perp}$) within each ATL08 segment can be obtained by averaging the across-track surface slope (with a 40-m interval) provided in the ATL06 product that is calculated for each pair using the first-photon-bias-corrected median heights [46]. Based on (5), the total surface slope (φ) is then calculated by using the along-track and across-track surface slope. With the resolved surface slope (φ), the corrected PNPS (N_{P08}-$h_s N_A$), and the fitting parameters obtained in Step (1), the expected signal photon number (N_s) can be calculated to correct the dead time effect using (15).

Step (3): We merge the captured photon events within seven successive along-track segments in the ATL06 products (over Antarctic plateau where the surface reflectance is given and the atmosphere is clear) and divide the captured photon event number by the corresponding laser shot number to generate the PNPS (N_{p06}) within 280-m segment. Such PNPS is used to calculate the calibration constant C_s for three selected laser tracks. Based on the ASR (ρ_{asr09}) and the dead time correction factors (D_c) in the ATL09 product, we can obtain the calibration constants as [44]

$$C_s = \frac{\pi R^2 E_p}{\gamma_r \gamma_d E_t A} \cdot C_F = \frac{\rho_{\text{asr09}}}{D_c N_{p06}}. \tag{16}$$

Step (4): The contribution of the two-way atmospheric transmittance primarily comprises absorption and scattering from the molecular, the cloud, and aerosol for the entire atmospheric column at 532 nm. The mean molecular transmission (T_m) only depends on the terrestrial altitude. The mean transmission of the cloud and aerosol can be derived from the optical depth (τ) in the ATL09 product with a 280-m interval. Thus, the two-way atmospheric transmittance can be approximately calculated as $T_a^2 = T_m^2 \exp(-2\tau)$. A closet temporal interpolation is implemented to provide the atmospheric transmittance for all segments (with a 100-m interval) in the used ATL08 dataset.

Step (5): The surface reflectance (ρ) is directly determined by substituting all abovementioned parameters into the radiometric model of the illuminated target in (8). An optimal threshold for the classification based on the surface reflectance histogram is determined by the Otsu thresholding method [58]. If the surface reflectance exceeds the threshold, the region is classified as snow-covered area; otherwise, it is classified as bare-land area.

IV. Results and Discussion

A. Radiometric Correction Model

Based on the simulation algorithm of photon-counting lidar data and the ATLAS specification parameters, we repeatedly (142 times) generate the actual returned signal photons, respond them as photon events, and calculate the corresponding PNPS (N_p) under the conditions of the total surface slopes (φ) ranging from 0° to 30° and the expected signal photon numbers (N_s) ranging from 0.1 to 10 counts. By resolving

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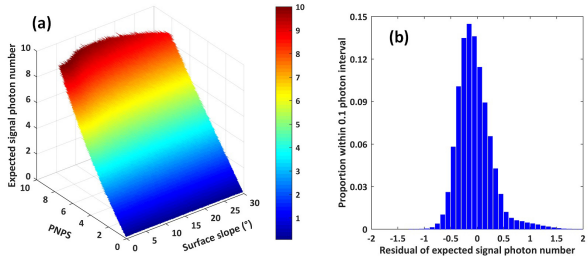


Fig. 3. (a) Relationship of the expected signal photon number (N_s), the PNPS (N_p), and total surface slope (φ). (b) Fitted residual of the expected signal photon number by a quadratic regression model. The surface slope in the quadratic regression model is units of degree. The intervals of the surface slope and expected signal photon number are 0.06 and 0.02, respectively. The fitted residuals are divided into 40 sections within a range from −2 to 2 and an interval of 0.1, and the proportion within each section is normalized by total segment numbers.

the PNPS for all segments, we provide the relationship of the expected signal photon number, the PNPS, and surface slope, as presented in Fig. 3(a).

According to the simulated data in Fig. 3(a), the quadratic regression model of the expected signal photon number is estimated as (17), and the corresponding histogram of the fitting residual is illustrated in Fig. 3(b). In Fig. 3(b), the coefficient of determination (r^2) is approximately 0.99 and the root mean square error (RMSE, δ_r) is 0.30 count. In addition, the fitting residual histogram generally satisfies a normal distribution with a mean of −0.05. The results indicate that the proposed quadratic regression model can well describe the relationship between the surface slope, the PNPS, and the expected signal photon number

$$N_s = \left(1.06 + \frac{7.14}{1000} N_p\right) N_p + \left(-0.07 + \frac{1.52}{1000}\varphi\right)\varphi + 0.58. \tag{17}$$

The quadratic regression model demonstrates that the PNPS has a greater impact on N_s than the surface slope. The impact of the surface slope could even be negligible for a larger PNPS but should be considerable for a smaller PNPS and a steeper slope. The derived radiometric correction model in (17) is very fundamental to all radiometric applications (e.g., the reflectance retrieval) using ICESat-2 data especially when the received signal is strong (i.e., N_s has several photons). In our previous study, when $N_s/M > 0.2$ counts (i.e., the expected signal photon number per detector), the dead time effect cannot be neglected, and the PNPS N_p will be less than actual N_s [59]. For ICESat-2/ATLAS with 16 detectors, when the total N_s is larger than three counts, the loss due to the dead time effect should be considered. Specifically, for strong beams of ICESat-2, the PNPS (N_p) from ice sheet, sea ice, and bare land with relatively high reflectance should be corrected and recovered to the expected signal photon number (N_s); otherwise, the derived radiometric results will be underestimated. For ocean subsurface applications, given that N_s reflected from water surfaces is normally less than four counts (mainly due to the wind speed) and the backscattered photon event number sharply decreases in water columns, the primary radiometric correction should be the after pulsing effect as proposed by Lu et al. [20].

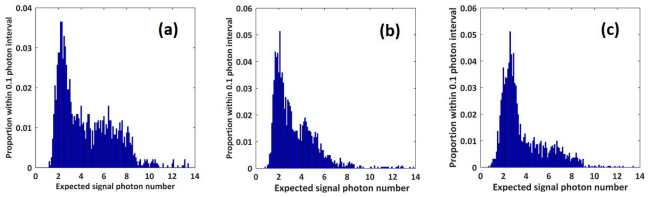


Fig. 4. Histogram distributions of expected signal photon numbers for three selected tracks in the study area. (a) GT1R, (b) GT2R, and (c) GT3R.

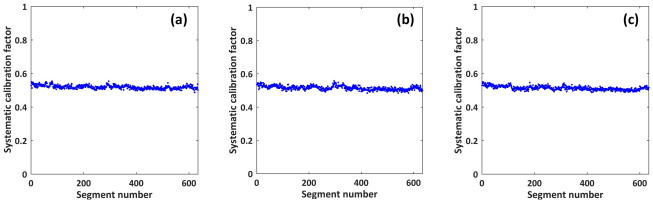


Fig. 5. Distributions of systematic calibration factors for three selected tracks of strong beams over Antarctic ice sheet. (a) GT1R, (b) GT2R, and (c) GT3R. The spatial resolution of each segment is 280 m in length.

B. Surface Reflectance Retrieval

1) Analysis on the Expected Signal PNPS: For three laser tracks in Fig. 1, using the corrected PNPS and the surface slope derived from the ATL08 and ATL06 product in Step (2) of Fig. 2 and the fitting model in (17), we can obtain the expected signal photon number (N_s) per laser shot. We divide N_s into 140 sections ranging from 0 to 14 counts and an interval of 0.1 and then calculate the proportion within each section. The histogram distributions of the expected signal photon number for three selected tracks are shown in Fig. 4. As displayed in Fig. 4, we observe that three histograms of the expected signal photon number (N_s) present bimodal patterns, which correspond to the signal photons from the bare land and snow in the study area, respectively.

2) Calculation on the Systematic Calibration Factor: For the given specification parameters of satellite lidars, the calibration constant in (5) mainly depends on the systematic calibration factor (C_F). We choose the ASR and the dead time correction factors (D_c) in the ATL09 product and calculate the PNPS (N_{p06}) within 280-m segment from the ATL06 product over Antarctic ice sheet for three selected tracks. According to the definition of the systematic calibration factor, we provide the systematic calibration factors (C_F) of the ICESat-2 using (7) and (16), as illustrated in Fig. 5.

All systematic calibration factors in Fig. 5 vary from 0.5 to 0.6, which are coincident with the settings after launch of the ICESat-2 [44]. The slight undulations of the systematic calibration factor are primarily rooted from the influence of the noise photons on the PNPS (i.e., some noise photons are not discarded and remained). We calculate the mean of systematic calibration factors for each laser track and notice that they are roundly equivalent to each other. Thus, the systematic calibration factors (C_F) are replaced by their means and prescribed to be 0.52 in this article.

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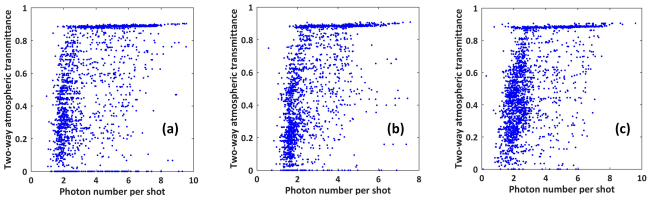


Fig. 6. Relationship between the PNPS and atmospheric transmittance for three ground tracks. (a) GT1R, (b) GT2R, and (c) GT3R.

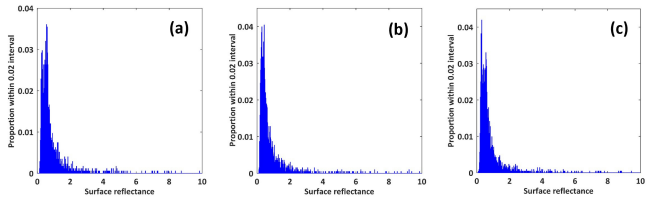


Fig. 7. Histogram distributions of surface reflectance for three ground tracks. (a) GT1R, (b) GT2R, and (c) GT3R.

3) Analysis on the Two-Way Atmospheric Transmittance: Based on the mean molecular transmission, the optical depth of the cloud and aerosol in the ATL09 product, and the closet temporal interpolation method, we calculate the two-way atmospheric transmittance and provide the relationships between the two-way atmospheric transmittance (T_a^2) and the corrected PNPS derived from ATL08 data for each 100-m segment, as presented in Fig. 6.

The results in Fig. 6 demonstrate that the atmospheric transmittance presents a tremendous variation for the smaller PNPS and even maintain at a lower level for the larger PNPS. It is an obviously abnormal phenomenon, which could be caused by the inaccurate calculation on the atmospheric optical depth. The optical depth in the ATL09 product is generated based on the Global Ozone Monitoring Experiment (GOME) monthly surface albedo and the ASR data [48]. However, the optical depth derived from the GOME-based data over the land area has a lower quality flag. It suggests that the optical depths over the study site may not be sufficiently credible.

4) Results of Surface Reflectance: By substituting all the abovementioned parameters into the radiometric model of the illuminated terrain in (8), we can obtain the surface reflectance (ρ) and divide them into 500 sections ranging from 0 to 10 and an interval of 0.02. The histogram distributions of the surface reflectance for the selected tracks are presented in Fig. 7. As illustrated from Fig. 7, we observe that surface reflectance may be larger than 1 for three selected tracks. It is an irrational situation introduced by the inaccurate atmospheric transmittance. A portion of the bare-land surface reflectance is overestimated, so that the histogram of bare land is heavily overlapped with snow, which introduces difficulty in the identification of the boundary between the bare land and snow and has strongly negative influence on their classifications.

In summary, the recovered N_s in Section IV-B1 and the calculated systematic calibration factor (C_F) in Section IV-B2 are generally much more credible than the derived surface reflectance (ρ) because the derived reflectance is extremely influenced by the atmospheric transmittance (T_a) that is not sufficiently credible. As shown in (8), the ASR (ASR $= \rho T_a^2$) is positively proportional to the product of the expected signal photon number and the calibration constant and thus is not influenced by the atmospheric transmittance. This implies that the ASR could also be applied and may be more preferable in the classification of the snow-covered area than the surface reflectance.

C. Land Cover Classification Results

1) Classification References and Threshold: The NDSI is employed to generate the referenced classifications of snow-covered land, which is treated as the truth values to verify the classification results from the ICESat-2 products. Based on the reflectance data in the third band and 11th band (r_3 and r_{11}) from Sentinel-2 L1C product, the NDSI can be calculated as

$$\text{NDSI} = \frac{r_3 - r_{11}}{r_3 + r_{11}} \tag{18}$$

where the spatial resolutions of the third band and 11th band are 10 and 20 m, respectively. The reflectance data in four pixels of the third band are merged and averaged to match the 20-m resolution of 11th band, so that the spatial resolution of the NDSI classification results is 20 m. Then, a threshold is determined according to the histogram distribution of the NDSI of four tiles in the study site. If the NDSI exceeds the threshold, the land cover of Sentinel-2 pixel is classified as snow; otherwise, it is classified as bare land [60]. However, the geolocation coordinates of the ATLAS laser footprints are not coincident with the Sentinel-2 pixels. In this study, if a laser footprint is located within the 20-m geolocated region of the pixel, the classification information of the laser footprint is assigned by that of the pixel. Meanwhile, each ATL08 segment comprises multiple laser footprints, whose classification is derived from the predominant land-cover types of laser footprints within the ATL08 segment. These classification results are treated as the referenced classifications in this article.

The classification threshold can be automatically determined by using the Otsu thresholding method [58]. It employs the surface reflectance histogram and searches for the maximization of between-class variance to separate the segmented classes. As the objects in the study site only are classified into two classes (bare land and snow), the between-class variance can be described by

$$\sigma^2(\kappa) = \eta_b(\kappa)\sigma_b^2(\kappa) + \eta_s(\kappa)\sigma_s^2(\kappa) \tag{19}$$

where η_b and η_s represent the probabilities for bare land and snow separated at a threshold κ, respectively, and σ_b and σ_s represent the corresponding variances of these two classes.

To validate the effect of the ASR on the classification of snow-covered areas, we additionally introduce the derived ASR from the expected signal photon number and the stored ASR in the ATL09 product to implement the classification experiments. The relationships between the classification thresholds and the between-class variances are illustrated in Fig. 8 for three ground tracks.

The relationships between thresholds and between-class variances in Fig. 8 present unimodal distributions. The optimal

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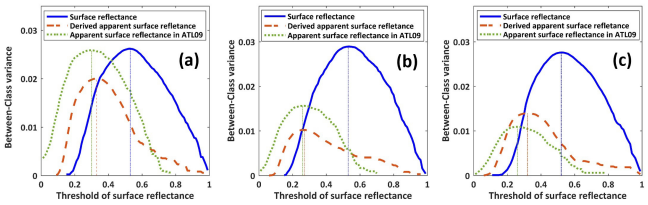


Fig. 8. Relationships between the thresholds and between-class variances for three ground tracks. (a) GT1R, (b) GT2R, and (c) GT3R. The blue, brown, and green vertical dashed lines represent the optimal thresholds for the derived surface reflectance, the derived ASR in this study, and the stored ASR in the ATL09 product, respectively.

TABLE II
OPTIMAL CLASSIFICATION THRESHOLDS FOR THREE GROUND TRACKS

Ground Track	Optimal Classification Threshold		
	Derived Surface Reflectance	Derived ASR	ASR in ATL09
GT1R	0.53	0.33	0.29
GT2R	0.53	0.27	0.26
GT3R	0.52	0.32	0.26

thresholds are located at the maximum of the between-class variances. We provide the optimal thresholds as listed in Table II. The optimal threshold in Table II obtained from the surface reflectance histogram is obviously larger than those from the derived ASR and the stored ASR in the ATL09 product. The optimal thresholds of ASRs are approaching to each other for three ground tracks.

2) Analysis on the Classification Results: Based on the proposed classification algorithm and classification thresholds, we implement the classification experiments over the snow-covered land in the study area by using the derived surface reflectance, the derived ASR, and the ASR in the ATL09 product. In addition, the ATL08 product provides a snow flag from the NOAA snow-ice daily product. All classification results are verified by the referenced classifications from Sentinel-2 (as truth values) for three selected tracks and illustrated in Fig. 9 (GT1R), Fig. 10 (GT2R), and Fig. 11 (GT3R), respectively.

The referenced classification results in Figs. 9–11 reveal that snow is primarily distributed on the upper surface of the ground. The classification results from the derived ASR are in the greatest accordance with the referenced classifications from Sentinel-2. It is unfortunately that the classification results from the surface reflectance and the NOAA snow-ice flag in the ATL08 product greatly deviate from the referenced classifications. The overestimation of the bare-land surface reflectance caused by the inaccurate atmospheric transmittance leads to such wrong classification of bare-land areas as snow-covered areas. The derived ASR is only related to the expected signal photon number (derived from ATL08 photons in the study area), system calibration constant (derived from ATL06 photons in Antarctic plateau), and the surface slope (derived from ATL08 photons and ATL06 cross-track slope in the study area), and these three parameters are more reliable than the

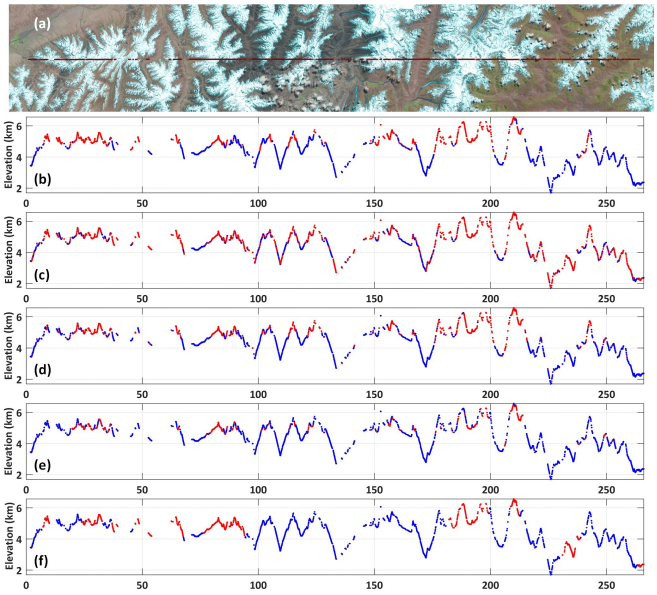


Fig. 9. Classification results using different classified data sources for GT1R in the study area. (a) Sentinel-2 imagery and the ground track of GT1R (the red line). From (b) to (f), the classification results are derived from the Sentinel-2 reflectance data (as truth values), the derived surface reflectance, the derived ASR, the stored ASR in ATL09 products, and NOAA snow-ice flag in ATL08 products, respectively. In (b)–(f), the snow-covered areas are marked with red points and the bare-land areas are marked with blue points.

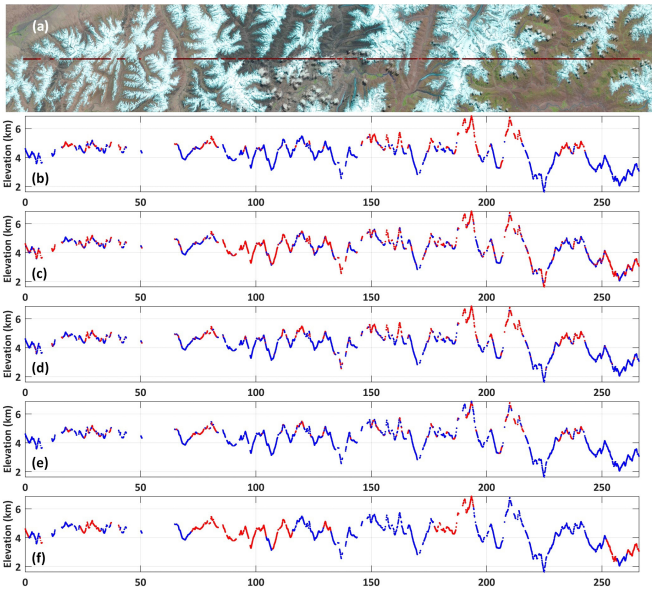


Fig. 10. Classification results using different classified data sources for GT2R in the study area. (a) Sentinel-2 imagery and the ground track of GT2R (the red line). From (b) to (f), the classification results are derived from the Sentinel-2 reflectance data (as truth values), the derived surface reflectance, the derived ASR, the stored ASR in ATL09 products, and NOAA snow-ice flag in ATL08 products, respectively. In (b)–(f), the snow-covered areas are marked with red points and the bare-land areas are marked with blue points.

atmospheric transmittance, which can explain that the derived ASR has the best results.

To quantitatively evaluate the classification results, we introduce two classification metrics including overall accuracy and the Kappa coefficient. The overall accuracy is the proportion of correctly classified samples to the total samples. The Kappa

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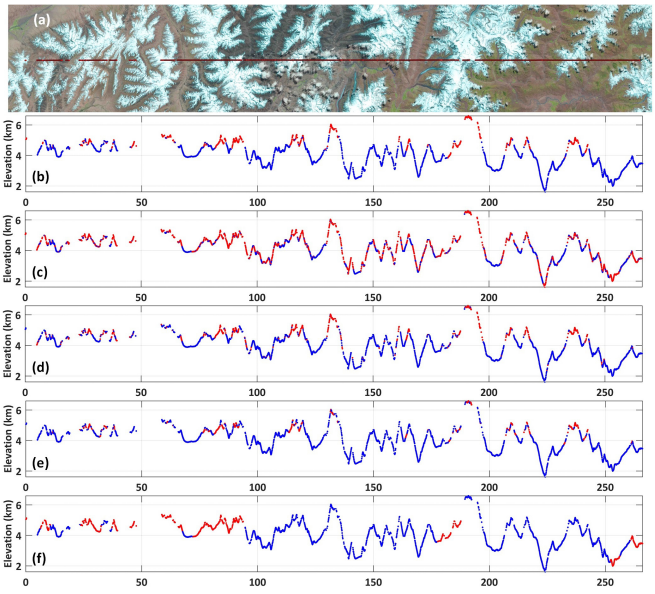


Fig. 11. Classification results using different classified data sources for GT3R in the study area. (a) Sentinel-2 imagery and the ground track of GT3R (the red line). From (b) to (f), the classification results are derived from the Sentinel-2 reflectance data (as truth values), the derived surface reflectance, the derived ASR, the stored ASR in ATL09 products, and NOAA snow-ice flag in ATL08 products, respectively. In (b)–(f), the snow-covered areas are marked with red points and the bare-land areas are marked with blue points.

coefficient measures the agreement between classification and reference. We define a 2×2 confusion matrix (M_{ij}), where diagonal elements of M_{11} and M_{22} represent the correct classifications of bare land and snow, and M_{12} and M_{21} represent the wrong classifications that bare land is classified as snow and snow is classified as bare land. According to the definitions of the overall accuracy and Kappa coefficient, we can express them with the confusion matrix elements as

$$\eta_{\mathrm{OA}} \frac{\sum_{i=1}^{2} M_{ii}}{N_F}$$
$$\eta_{\mathrm{Kap}} = \frac{N_F \sum_{i=1}^{2} M_{ii} - \sum_{k=1}^{2} \left(\sum_{j=1}^{2} M_{kj} \cdot \sum_{i=1}^{2} M_{ik} \right)}{N_F^2 - \sum_{k=1}^{2} \left(\sum_{j=1}^{2} M_{kj} \cdot \sum_{i=1}^{2} M_{ik} \right)} \tag{20}$$

where η_{OA} and η_{Kap} denote the overall accuracy and Kappa coefficient, and N_F is the total number of the classified samples. The evaluations of the classification results for three tracks are shown in Table III.

The classification metrics in Table III indicate a demonstration that the overall accuracies and Kappa coefficients obtained from the derived surface reflectance and NOAA product are relatively inferior. It again indicates that the inaccurate atmospheric transmittance is the principal factor resulting in the erroneous classifications from the surface reflectance. Fortunately, the classification quality has a rapid improvement by only employing the derived ASR as the classified data. Based on the derived ASR, we can get the best classification results with the overall accuracies of 82.84%, 85.38%, and 88.80% and Kappa coefficients of 0.64, 0.65, and 0.69 for three laser tracks. The classification accuracies

TABLE III
EVALUATION OF THE CLASSIFICATION RESULTS FOR THREE GROUND TRACKS

Ground Track	Classified Data Source	Classification Evaluation Metrics	
		Overall Accuracy	Kappa Coefficient
GT1R	Derived Surface Reflectance	56.62%	0.18
	Derived ASR	82.84%	0.64
	ASR in ATL09	75.56%	0.46
	NOAA Snow-ice in ATL08	67.40%	0.31
GT2R	Derived Surface Reflectance	44.91%	-0.09
	Derived ASR	85.38%	0.65
	ASR in ATL09	82.01%	0.53
	NOAA Snow-ice in ATL08	55.87%	0.00
GT3R	Derived Surface Reflectance	57.26%	0.20
	Derived ASR	88.80%	0.69
	ASR in ATL09	86.80%	0.58
	NOAA Snow-ice in ATL08	71.85%	0.25

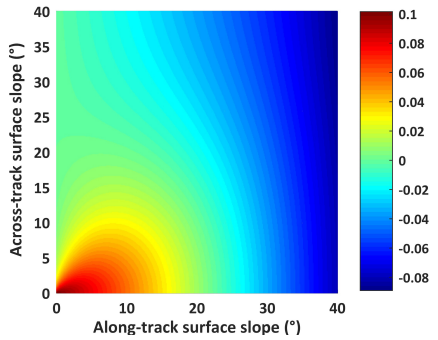


Fig. 12. Differences of the expected signal photon number between laser pointing angles of 1.5° and 0° for different surface slopes.

from the ASR in the ATL09 product have small decrease with overall accuracies of 75.56%, 82.01%, and 86.80% and Kappa coefficients of 0.46, 0.53, and 0.58. Such deviations undoubtedly verify that the radiometric correction in (17) on the expected signal photon number is meaningful in radiometric applications for ICESat-2 data.

D. Influence of Laser Pointing Angle on Correcting N_s

In the abovementioned investigations on the laser incident angle φ, i.e., (5), we ignore the contributions from the laser pointing angle but only consider the total surface slope. We investigate the specification and configuration of ICESat-2, which indicates that the maximum laser pointing angle is approximately 1.5°. Accordingly, under the conditions of laser pointing angles of 1.5° and 0°, we separately calculate the expected signal photon numbers N_s for different surface slopes by using (4) and (17). Then, we calculate the difference between the results of N_s under the conditions of laser pointing angles of 1.5° and 0° and illustrate this difference in Fig. 12.

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The results in Fig. 12 indicate that the maximal difference of N_s is within 0.1 count, which corresponds to the smaller surface slopes. Fortunately, such differences have a relatively weak impact on the radiometric correction as the difference is much smaller than the fitted residual, i.e., 0.3 count, as shown in Fig. 3(b).

V. Conclusion

The radiometric information of the illuminated target can be obtained according to the returned signal strength, the surface slope, the systematic calibration factor, and two-way atmospheric transmittance. We implement the radiometric correction empirical model for satellite photon-counting lidars to remove the influence of the nonlinear response and dead time of detectors, i.e., recover the expected signal photon number N_s from captured PNPS. The model in (15) is generally useful for spaceborne photon-counting lidars with PMTs when measuring ground surfaces. However, as the fitting parameters are not only related to environmental parameters but also system parameters of a given photon-counting lidar, the fitted parameters in (17) is only specific to ICESat-2/ATLAS.

The radiometric correction is very essential to radiometric applications for photon-counting lidars, and for strong beams of ICESat-2, this correction should be conducted when using data captured on ice sheet, sea ice, and bare land with relatively high reflectance (e.g., the used area in this study). The radiometric correction results over the study site demonstrate that the expected signal photon number and systematic calibration factor can be derived accurately, but the atmospheric transmittance root from the ATL09 product is sometimes inaccurate. The land surface reflectance is usually overestimated due to inaccurate atmospheric transmittance especially when the PNPS is relatively small.

With the above achievement, we further carry out the classification experiments by employing the surface reflectance and the ASR data based on the Otsu thresholding method. By comparing the classification results with the referenced classifications obtained from the Sentinel-2 L1C product for three strong beam tracks, we find that the classification results used the surface reflectance and NOAA snow-ice flag in ATL08 products are inferior with lower overall accuracies and Kappa coefficients. The main reason is that the overestimation of the bare-land surface reflectance (caused by the inaccurate atmospheric transmittance), which makes some bare-land areas wrongly identified as snow-covered areas. Based on the ASR derived in this article, the highest overall accuracy and Kappa coefficient are achieved with 88.80% and 0.69. It suggests that we should employ the ASR as classified data to improve the classification accuracy, which may be probably indeed for other radiometric applications when using ICESat-2 data.

Acknowledgment

The authors would like to acknowledge NASA's National Snow and Ice Data Center for providing the ICESat-2 data and the Sentinel Scientific Data Hub of the European Space Agency for providing the Sentinel-2 data. They also thank the editor and anonymous reviewers for reviewing our article.

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